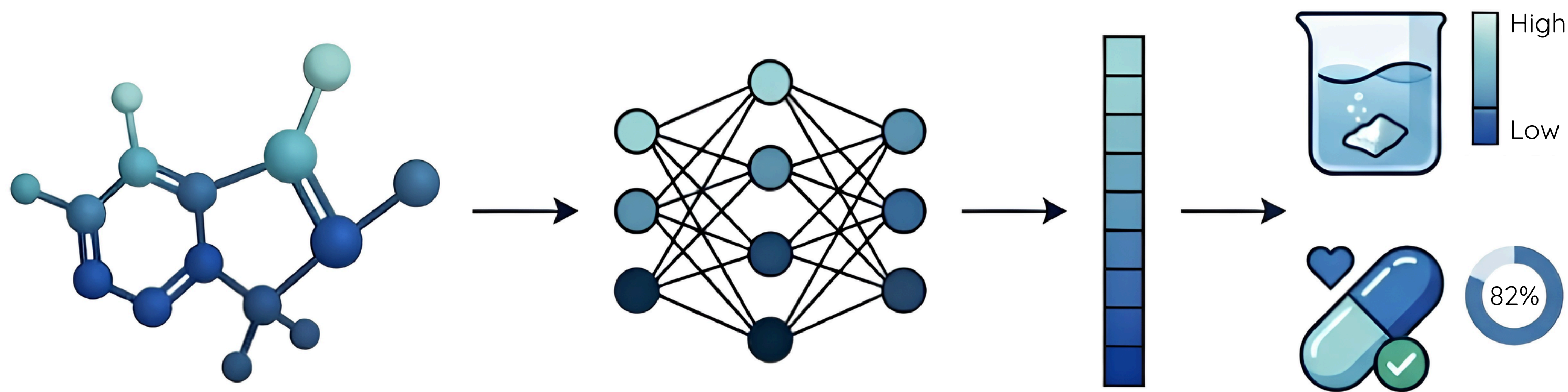
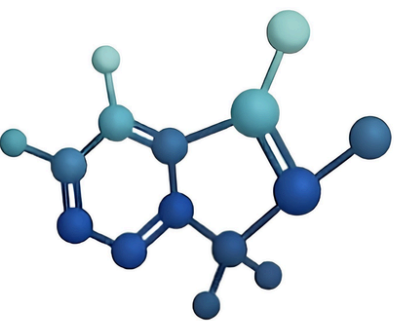
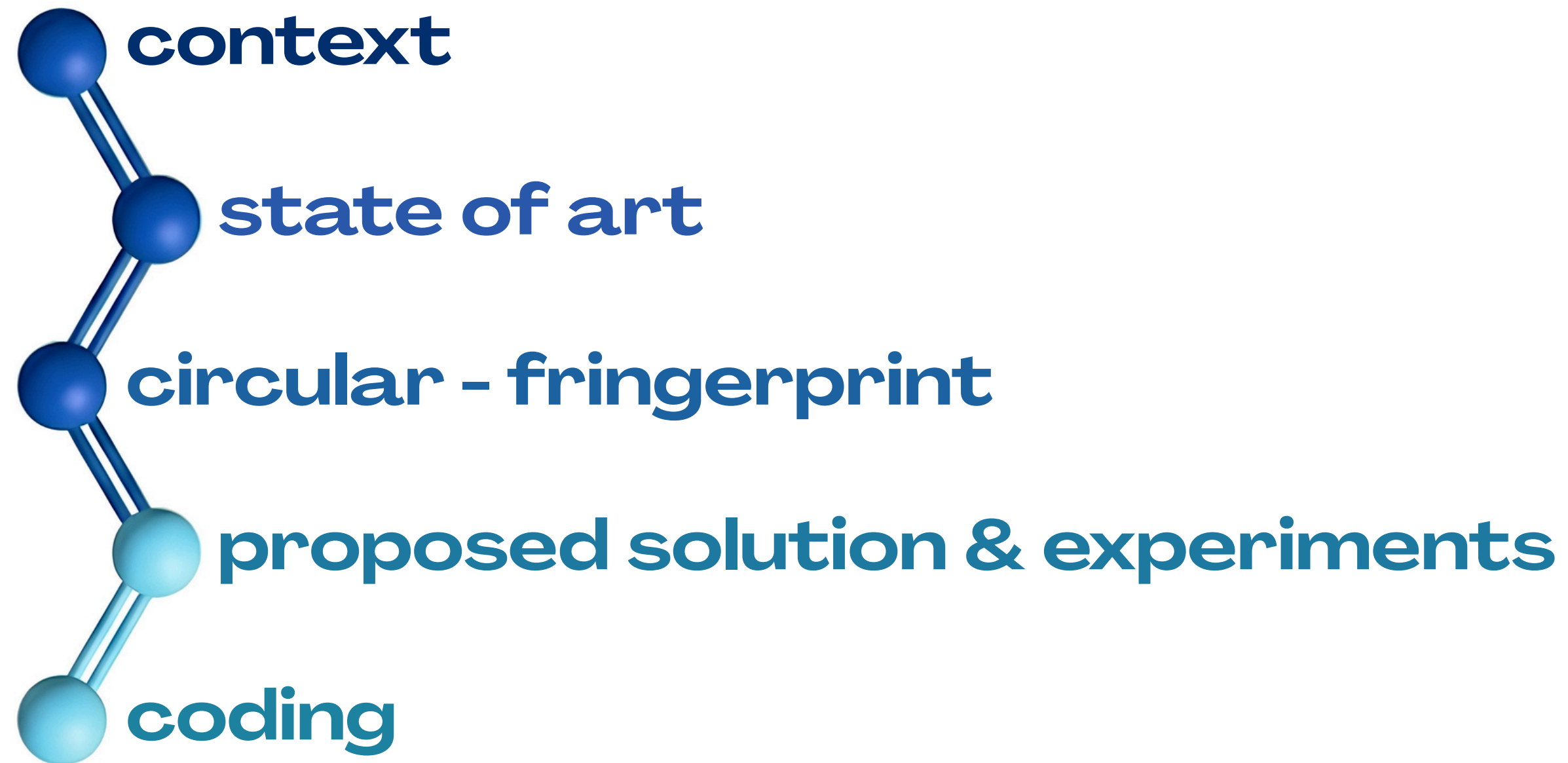


Convolutional Networks on Graphs for Learning Molecular Fingerprints

D. Duvenaud, D. Maclaurin, J. Aguilera-Iparraguirre, R. Gómez-Bombarelli, T. Hirzel, A. Aspuru-Guzik, R. Adams (2015)



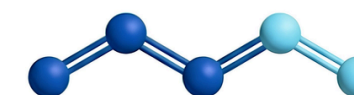
index



context

- **Molecular structures display fully arbitrary complexity:** they can range from just a few atoms to several hundred, and may take linear, branched, or cyclic forms.
- **Classical machine learning models, require fixed-dimension input vectors,** which are inherently incompatible with the variable-size, graph-like nature of molecular representations.
- This creates a **fundamental mismatch** between the variable nature of molecular graphs and the rigid vector-based architecture of traditional neural networks.

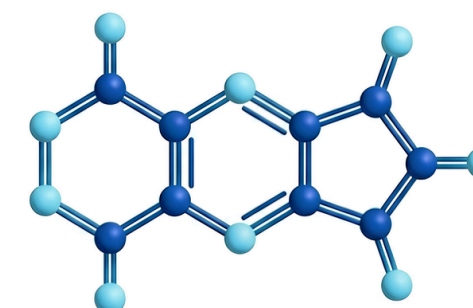
Linear
(Small, 4 atoms)



Branched
(Mid-sized, 15 atoms)



Fused Cyclic
(Large, 30+ atoms)



state of the art

- Before this paper, molecular property prediction relied on a rigid pipeline:

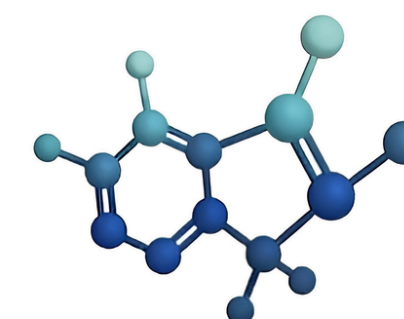
1. **External Feature Extraction**

the molecule was converted into a deterministic fixed-size vector (typically 1024 or 2048 elements) using external cheminformatics software and predefined algorithms.

2. **Standard Prediction**

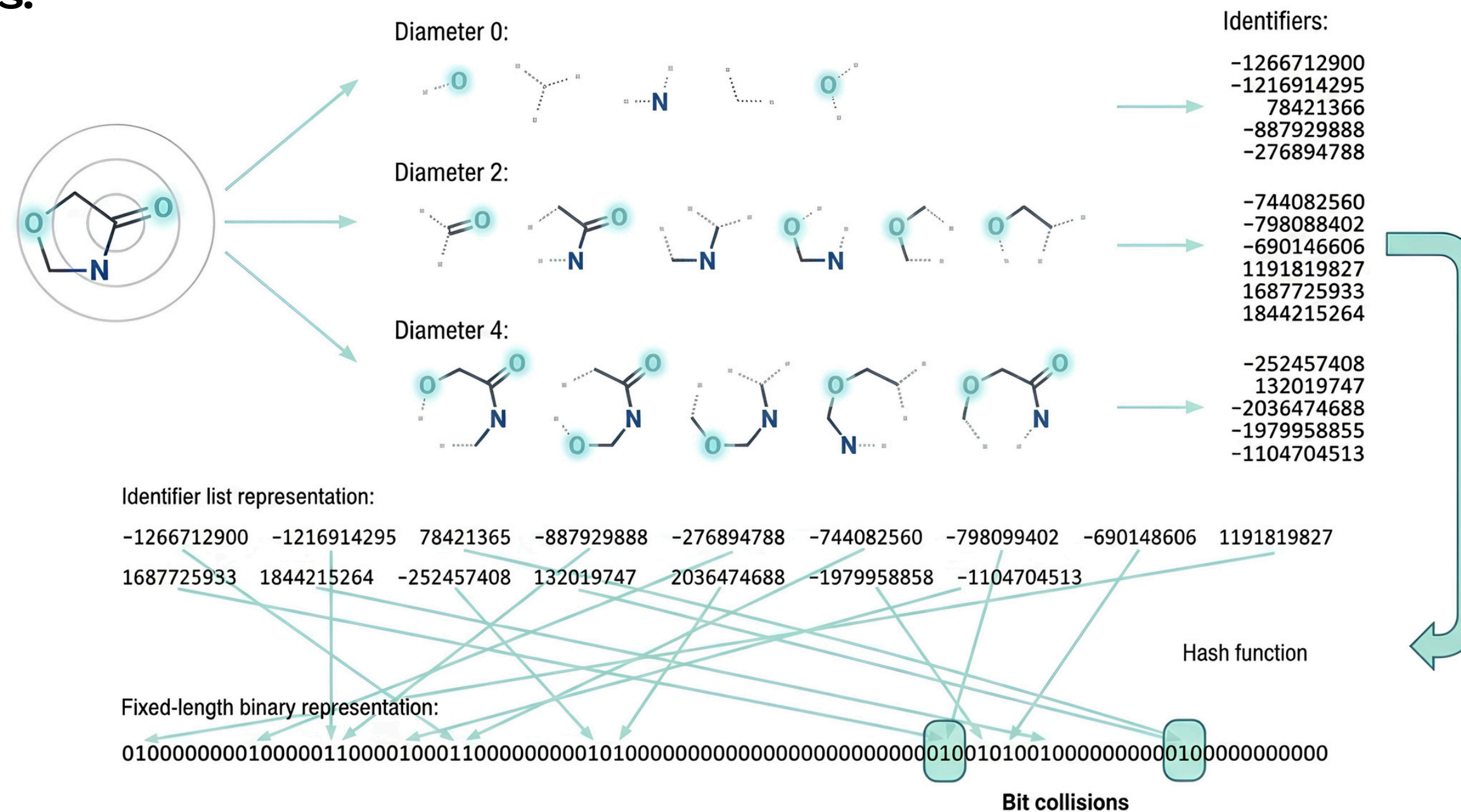
the resulting feature vector, static and non-modifiable, was then fed into a fully connected neural network or other classical machine-learning models to perform the prediction.

- The fundamental **limitation** was that the features were not learned from data but generated through predefined rules. The extraction process remained rigid, non-adaptable, and unable to specialize to the specific task.



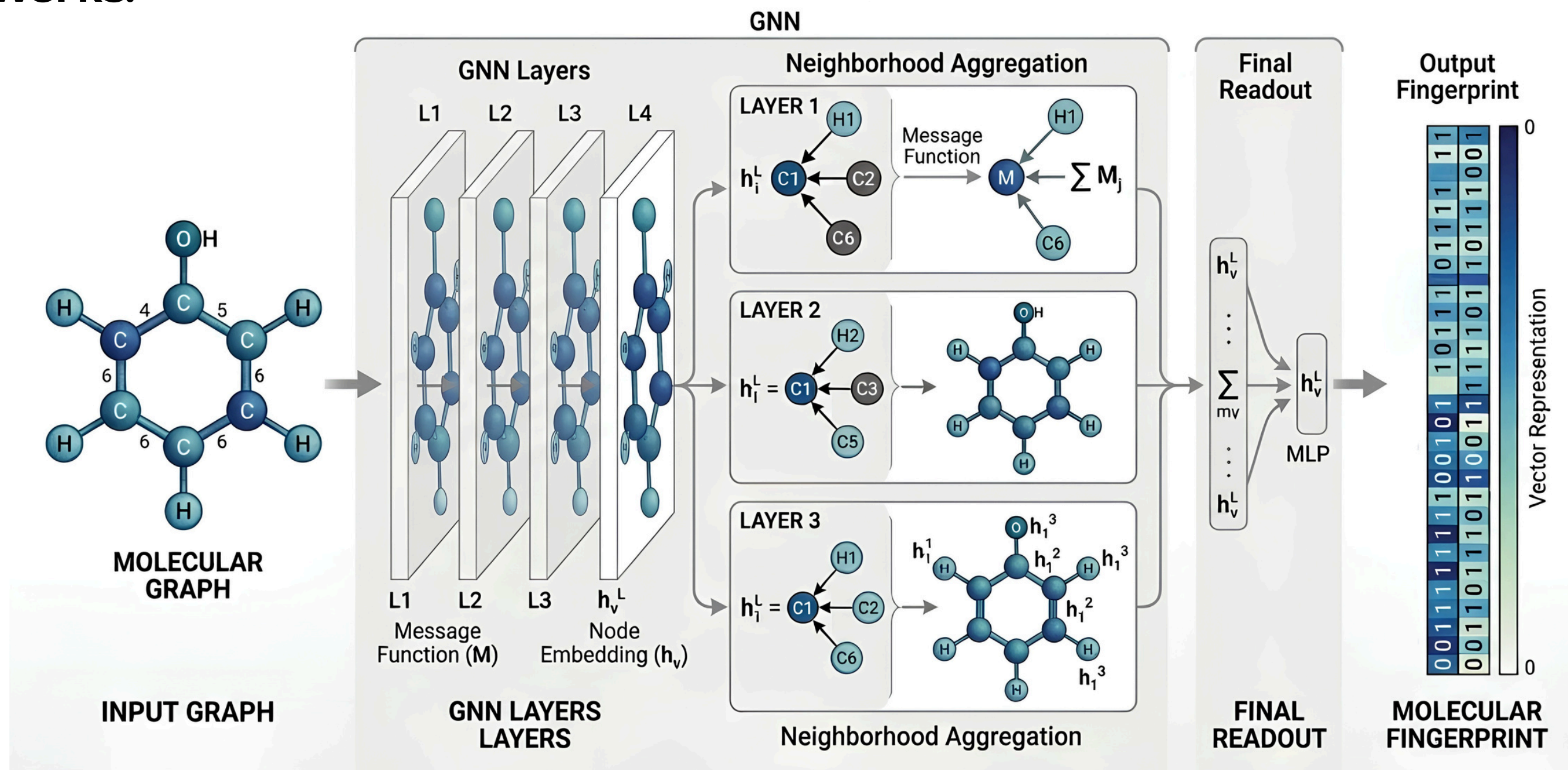
circular - fingerprint

- A family of algorithms that represents a molecule by identifying its small chemical substructures.
- **How it works:**



proposed solution

- In this work, the authors propose to completely replace the static fingerprinting stage with an end-to-end approach.
- **How it works:**



proposed solution - advantages

- The end-to-end approach is not only conceptually more elegant, but also delivers concrete benefits in terms of accuracy, computational efficiency, and interpretability.

advantage 1

- The model learns which molecular features truly matter for the task.
- Performance matches or even surpasses circular fingerprints.

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advantage 1

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advantage 2

- Neural fingerprints are compact because they encode only the truly relevant features.
- This leads to clear benefits:
 - lower computational cost
 - reduced need for regularization.

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advantage 1

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- Performance matches or even surpasses circular fingerprints.

advantage 2

- Neural fingerprints are compact because they encode only the truly relevant features.
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 - lower computational cost
 - reduced need for regularization.

advantage 3

- In circular fingerprints, each molecular fragment is treated as an isolated index, with no notion of similarity.
- Neural fingerprints rely on continuous and shared features.

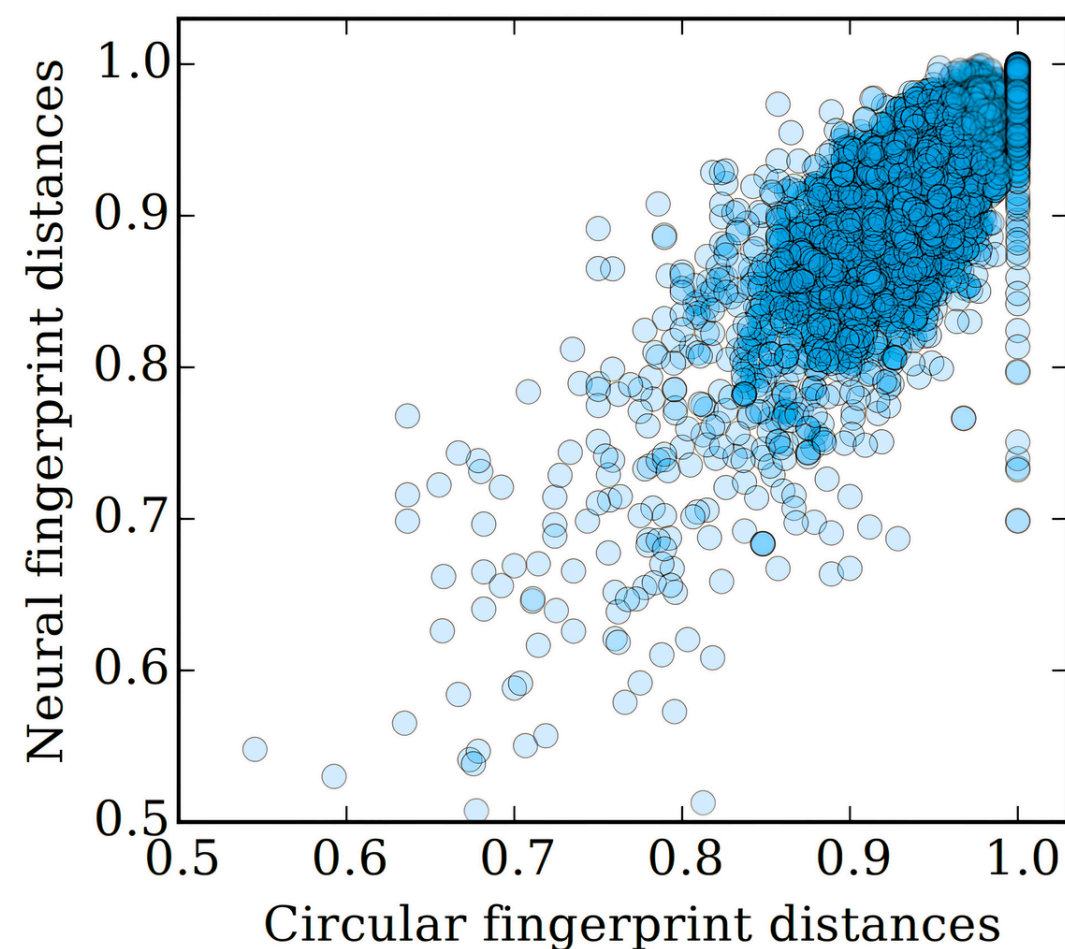
proposed solution – how to implement it

- The fundamental idea is to convert every discrete step of traditional circular fingerprints into a continuous and differentiable operation.

operation	circular fingerprint	neural graph fingerprint
Hashing	$\text{hash}(v)$	neural layer: $\sigma(v \cdot H)$
Indexing	$\text{mod}(r, S)$	$\text{softmax}(r \cdot W)$
Canonicalization	neighbor ordering	permutation-invariant

experiment n. 1 - generalization test

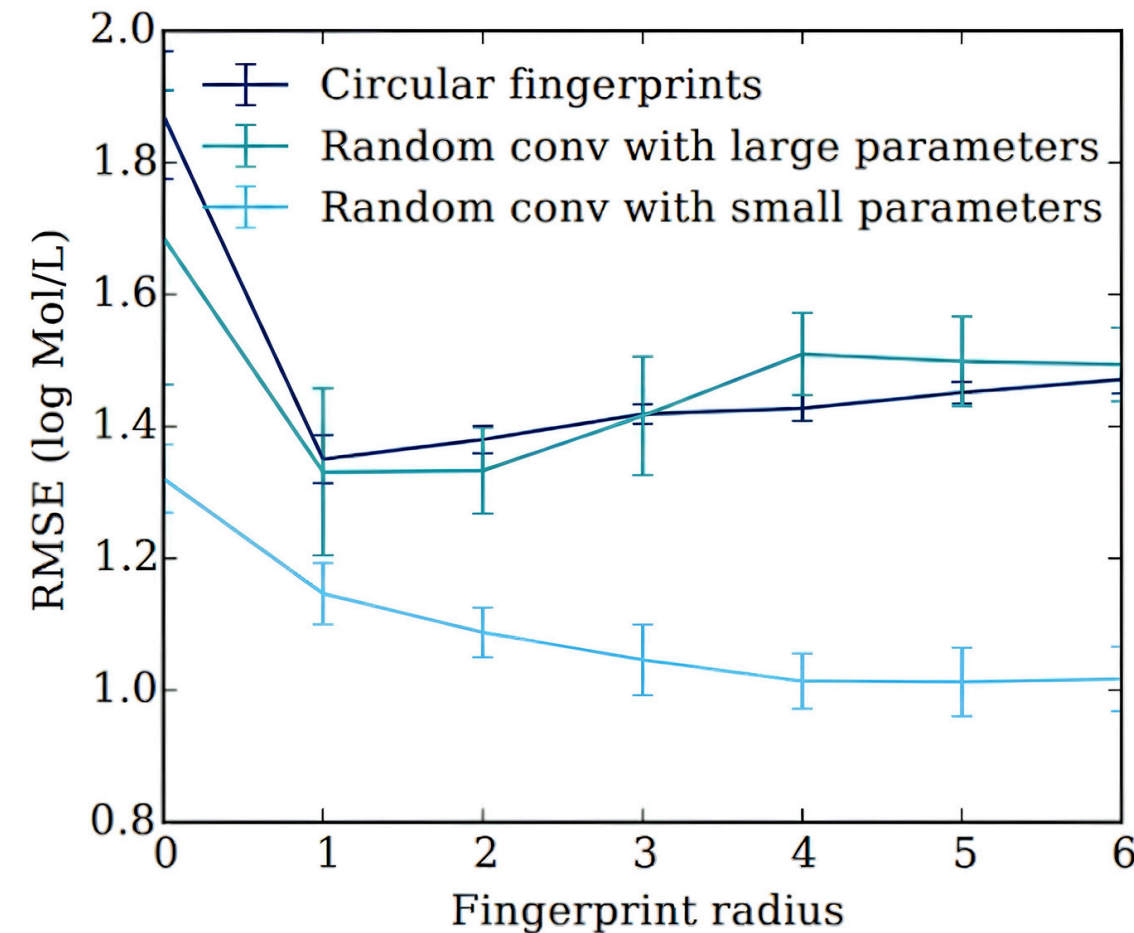
- The experiment investigates whether molecules considered similar according to circular fingerprints are also similar according to neural fingerprints, using randomly initialized networks with large weights.
 - 2048-dimensional vectors are computed for solubility datasets and compared using the Tanimoto distance: $d(x, y) = 1 - \frac{\sum_i \min(x_i, y_i)}{\sum_i \max(x_i, y_i)}$



$r = 0.823$

experiment n. 2 - predictive capabilities

- The authors compare the predictive capabilities of the fingerprints using a simple linear regression to predict solubility under three conditions:
 - Circular fingerprints
 - Neural fingerprints with large random weights
 - Neural fingerprints with small random weights



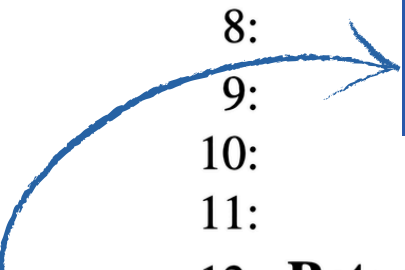
coding – problem 1

- **Structural Discrepancy**

- The comparative analysis highlights a key structural divergence between the theoretical formulation and the official codebase:
 - The implementation separates the transformation of the central atom from that of its neighbors, using two distinct matrices.

Algorithm 2 Neural graph fingerprints

```
1: Input: molecule, radius  $R$ , hidden weights  $H_1^1 \dots H_R^5$ , output weights  $W_1 \dots W_R$ 
2: Initialize: fingerprint vector  $\mathbf{f} \leftarrow \mathbf{0}_S$ 
3: for each atom  $a$  in molecule
4:    $\mathbf{r}_a \leftarrow g(a)$   $\triangleright$  lookup atom features
5: for  $L = 1$  to  $R$   $\triangleright$  for each layer
6:   for each atom  $a$  in molecule
7:      $\mathbf{r}_1 \dots \mathbf{r}_N = \text{neighbors}(a)$ 
8:      $\mathbf{v} \leftarrow \mathbf{r}_a + \sum_{i=1}^N \mathbf{r}_i$   $\triangleright$  sum
9:      $\mathbf{r}_a \leftarrow \sigma(\mathbf{v} H_L^N)$   $\triangleright$  smooth function
10:     $\mathbf{i} \leftarrow \text{softmax}(\mathbf{r}_a W_L)$   $\triangleright$  sparsify
11:     $\mathbf{f} \leftarrow \mathbf{f} + \mathbf{i}$   $\triangleright$  add to fingerprint
12: Return: real-valued vector  $\mathbf{f}$ 
```



python

```
self_activations = np.dot(atom_features, layer_self_weights)
neighbour_activations = matmult_neighbors(
    array_rep, atom_features, bond_features, get_weights_func)

total_activations = neighbour_activations + self_activations + layer_bias
```

coding - problem 2

- **Batch Normalization**

- The implemented Batch Normalization completely omits the learnable parameters γ and β .
- Without them, the network loses the ability to modulate or undo the normalization, effectively reducing it to a standardization mechanism inspired by Batch Normalization.

```
x_normalized = (x - batch_mean) / sqrt(batch_variance +  $\epsilon$ )  
y =  $\gamma$  · x_normalized +  $\beta$ 
```

what does the code:

```
python  
  
def batch_normalize(activations):  
    mbmean = np.mean(activations, axis=0, keepdims=True)  
    return (activations - mbmean) / (np.std(activations, axis=0, keepdims=True) + 1)
```

coding – problem 3

- **Silent Bug in the pipeline**

- The degree of an atom is encoded using a fixed-category one-hot encoding, rigidly limited to values between 0 and 5.
- Molecules containing atoms with a degree ≥ 6 pass the initial checks but cause a critical model crash in later stages.

```
python
```

```
one_of_k_encoding(atom.GetDegree(), [0, 1, 2, 3, 4, 5])
```

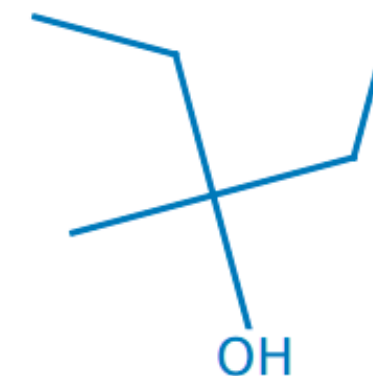
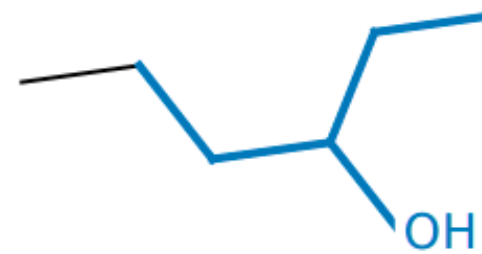
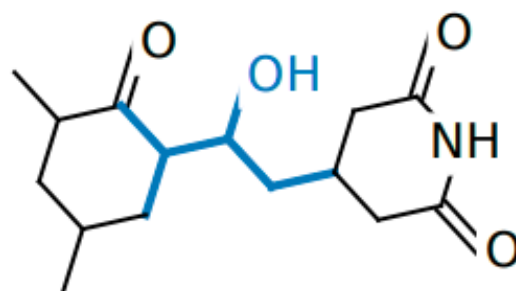
```
python
```

```
def one_of_k_encoding(x, allowable_set):  
    if x not in allowable_set:  
        raise Exception(...)
```

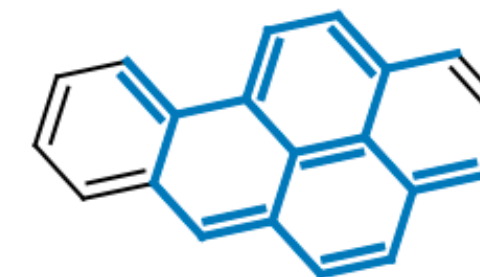
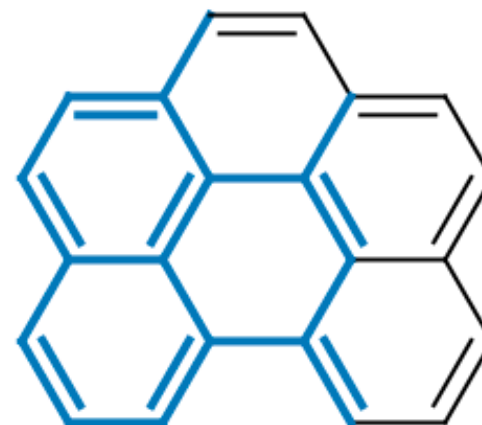
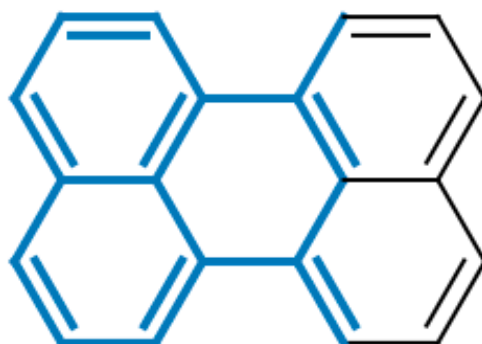
coding - interpretability (1)

- Solubility features

Fragments most
activated by
pro-solubility
feature



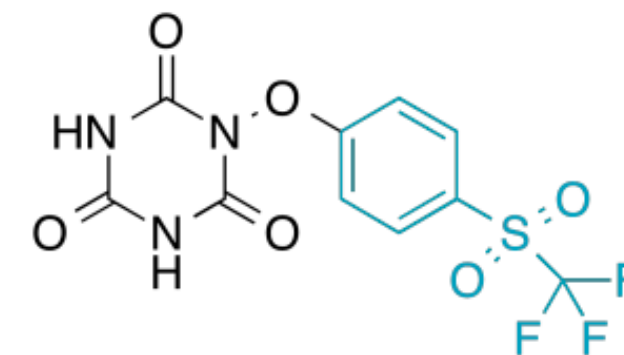
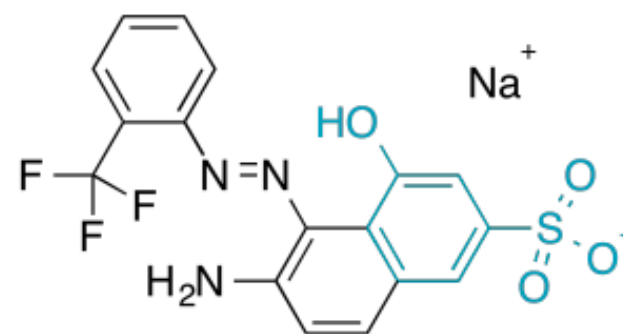
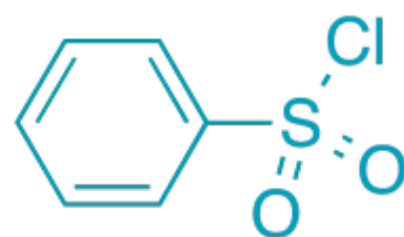
Fragments most
activated by
anti-solubility
feature



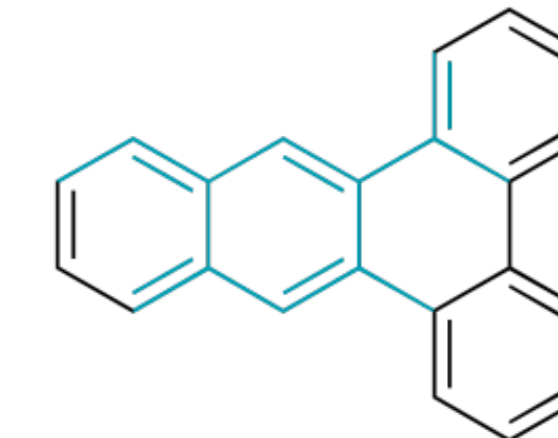
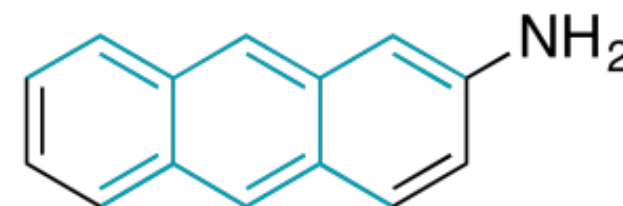
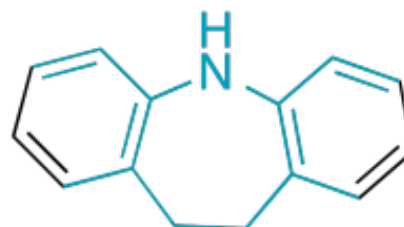
coding - interpretability (2)

- Toxicity features

Fragments most
activated by
toxicity feature
on SR-MMP
dataset



Fragments most
activated by
toxicity feature
on NR-AHR
dataset



coding – predictive performance

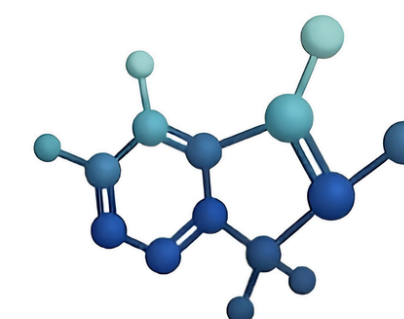
- The authors conduct a **series of experiments** to directly compare their predictive performance. In particular, the performance of the fingerprints is evaluated under two conditions:
 - using a linear model
 - using a neural network with a single hidden layer

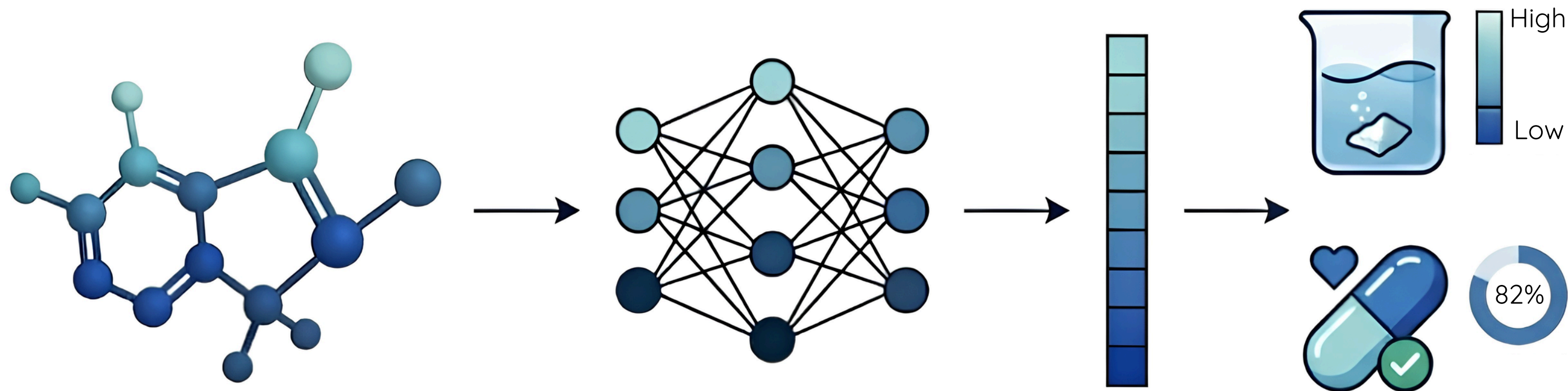
Dataset Units	Solubility log Mol/L	Drug efficacy EC ₅₀ in nM	Photovoltaic efficiency percent
Predict mean	4.29 ± 0.40	1.47 ± 0.07	6.40 ± 0.09
Circular FPs + linear layer	1.71 ± 0.13	1.13 ± 0.03	2.63 ± 0.09
Circular FPs + neural net	1.40 ± 0.13	1.36 ± 0.10	2.00 ± 0.09
Neural FPs + linear layer	0.77 ± 0.11	1.15 ± 0.02	2.58 ± 0.18
Neural FPs + neural net	0.52 ± 0.07	1.16 ± 0.03	1.43 ± 0.09

proposed solution – limitation

- **Computational cost:** neural graph fingerprints have the same asymptotic complexity as circular fingerprints, but introduce additional computational costs due to the neural operations.
- **Limited layer complexity:** the architecture chosen is deliberately minimal: each message-passing step uses a single neural layer.
- **Limited information propagation:** neural graph fingerprints rely on local message passing: each atom communicates only with its immediate neighbors, then with the neighbors of those neighbors, and so on. This approach scales well for organic molecules.
- **Inability to distinguish stereoisomers:** in the version presented in the paper, neural graph fingerprints cannot distinguish between:
 - enantiomers (mirror-image molecules),
 - cis/trans isomers (different double-bond geometries).

Stereochemistry requires additional information that is not included in the 2D molecular graph used by the model.





Thank you for your attention